MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP(HONOURS)REGULAREXAMINATIONMarch2026

Fourth SEMESTER

SkillEnhancementCourse(SEC)-MG4SECFSQ201-FIRST-AID,FIRE SAFETYAND DISASTER MANAGEMENT

(2024ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest(I) and Appreciation(Ap)

Studentsshouldattemptatleastonequestionfromeachcourseoutcometoenhancetheiroverall outcome attainability.

Part A

MCQ

Answerallquestions

Eachquestioncarries**1**marks

1. **b)** Local disaster management committees
2. **a)** National Disaster Response Fund
3. **b)** Physical phenomenon posing threat to life and property
4. **b)** Time-sensitive actions to save life and property
5. **b)** Hazard × vulnerability = disaster
6. **c)** National Executive Committee
7. **d)** Agricultural development
8. **b)** Apply further dressing over the initial one
9. **b)** 3 to 5 minutes or more
10. **a)** All human body fluids and secretions
11. **c)** A collarbone break from falling on an outstretched hand
12. **b)** To prevent death even if the injury itself isn't fatal
13. **b)** Call emergency services
14. **b)** Partial and Complete
15. **b)** To provide quick, effective solutions for immediate treatment and stabilization
16. **b)** Fire detection
17. **b)** Chemical reaction
18. **d)** Water
19. **c)** To verify fire resistance and safety measures
20. **c)** 15–20°F per minute

PartB

ShortAnswer

Answerany**5**questions

Eachquestioncarries**4**marks

21. Definemitigationandexplainwhyitisconsideredthecornerstoneofdisaster management.

Mitigation refers to long-term measures taken to reduce or eliminate the risk and impact of disasters before they occur.

- It focuses on risk reduction rather than response after a disaster.
- Includes structural measures such as earthquake-resistant buildings, flood control dams and cyclone shelters.
- Includes non-structural measures like policies, land-use planning and building regulations.
- Reduces loss of life, property damage and economic loss.
- Strengthens community resilience and preparedness.
- Therefore it is considered the cornerstone of disaster management.

22. Summarizetheimportanceofcommunityawarenessandeducationprogramsin risk reduction.

Community awareness programs help people understand disaster risks and preparedness methods.

- Educates people about early warning signals.
- Promotes preparedness measures such as evacuation plans.
- Helps people learn first aid and emergency response skills.
- Encourages community participation in disaster management activities.
- Reduces panic and confusion during emergencies.
- Improves coordination between authorities and the public.
- Helps reduce casualties and property damage.

23. Givethesignalsofapossiblyvenomoussnakebite.

A venomous snakebite may produce several visible symptoms that indicate the presence of venom in the body. Recognizing these signs quickly helps in providing proper medical treatment.

- Two puncture marks (fang marks) at the bite site
- Pain and swelling around the wound
- Bleeding or discoloration of skin
- Nausea, vomiting or dizziness
- Difficulty in breathing or swallowing

- Blurred vision or drooping eyelids
- Weakness or paralysis in severe cases

24. Explainwhyarescuershouldallowforcomplete"chestrecoil"aftereach compression and why a firm surface is necessary for infants.

During CPR, proper technique is essential to ensure effective blood circulation in the body. Chest recoil and a firm surface help make compressions more effective.

Chest recoil

- Allows the heart to refill with blood between compressions
- Improves blood circulation to the brain and vital organs
- Prevents reduced blood flow due to continuous pressure

Firm surface for infants

- Helps achieve adequate compression depth
- Soft surfaces reduce the effectiveness of compressions
- Ensures proper transmission of force to the heart

25. Discusshowafireextinguisherworkstostopafire.

Fire extinguishers are designed to control small fires by interrupting the process of combustion. They work by removing one or more elements of the fire triangle.

- Cooling: Water extinguishers lower the temperature below the ignition point.
- Smothering: Foam or CO_2 removes oxygen from the fire.
- Chemical interruption: Dry powder stops the chemical chain reaction of fire.
- Prevents the spread of fire.
- Provides time for safe evacuation and fire control.

26. Listanyfivecommonapplicationsorplaceswherefireblanketsareused.

Fire blankets are safety devices used to extinguish small fires by covering them and cutting off oxygen supply. They are commonly used in places where fire hazards may occur.

- Kitchens, especially for cooking oil fires
- Laboratories
- Industrial workshops
- Hospitals and first aid rooms
- Schools and colleges
- Fire safety equipment kits in buildings

- For extinguishing burning clothes

27. Listandexplainanyfiveimportantmaintenancerequirementsoffiresprinkler systems.

Regular maintenance of fire sprinkler systems is necessary to ensure that they function properly during a fire emergency. Proper inspection and testing improve reliability.

- Regular inspection of sprinkler heads
- Ensure sprinklers are not blocked by dust or objects
- Check water supply and pressure levels
- Test control valves and alarm systems periodically
- Conduct system flushing and testing
- Inspect for leaks, corrosion or damage
- Maintain proper maintenance and inspection records

PartC

Essay

Answerany**1**questions

Eachquestioncarries**10**marks

28. Writeanessayonfirstaidmeasuresforpoisoninganddrowning.

Poisoning and drowning are common emergencies that require immediate first aid. Quick and correct action can save lives before professional medical help arrives.

First Aid for Poisoning

Poisoning occurs when harmful substances enter the body through ingestion, inhalation, injection or skin contact.

- Identify the type of poison if possible
- Call emergency medical services immediately
- Remove the victim from the source of poison
- Keep the person calm and comfortable
- Move the victim to fresh air if poison is inhaled
- Do not induce vomiting unless instructed by medical professionals
- Wash the skin with clean water if poison contacts skin
- Keep the poison container or sample for doctors

First Aid for Drowning

Drowning occurs when breathing is impaired because water enters the airway.

- Remove the victim from water as quickly and safely as possible
- Check breathing and pulse

- Start CPR immediately if the person is not breathing
- Loosen tight clothing and keep the victim warm
- Place in recovery position if breathing resumes
- Seek immediate medical assistance
- Monitor the victim until professional help arrives

29. DifferentiatebetweenIntentionalandUnintentionalfire.Giveexamples.

Fires can be broadly classified into intentional and unintentional fires depending on how they are caused. Understanding this difference helps in prevention and investigation.

Intentional Fire

Intentional fire is deliberately started by a person with a specific purpose.

Examples include

- Arson for insurance fraud
- Riots and vandalism
- Terrorist attacks
- Revenge or sabotage

These fires are usually planned and may involve accelerants, causing serious damage to life and property.

Unintentional Fire

Unintentional fire occurs accidentally due to negligence or technical faults.

Examples include

- Electrical short circuits
- Gas leaks and explosions
- Unattended cooking
- Improper handling of flammable substances
- Faulty electrical appliances

Such fires are not deliberate and often result from human error or equipment failure.